P101 · Unit 9 · Quick reference

Physics, space & scientific thinking
Travelling waves and superposition
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Recall **Space and time**	λ: spatial repetition / m. T: time repetition / s. $f = 1 / T$ / Hz. Propagation speed $v_w = f\lambda$.
Recall **Phase scales**	$k = 2\pi / \lambda$ / rad m^{-1}; $\omega = 2\pi f$ / rad s^{-1}. kx, ωt, ϕ: rad.
Recall **Travelling profile**	$y = A\cos(kx - \omega t + \phi)$: rightward for $k, \omega > 0$. Plus time term: leftward. Speed ω / k; local material motion differs.
Understand **Linear superposition**	$y = y_1 + y_2$, at the same position/time and common zero. Small disturbances in a linear model; add signed deviations.
Consult **Equal-wave phase sum**	Same direction, frequency, wavelength, amplitude: $y = 2A\cos(\delta / 2)\cos(\theta + \delta / 2)$, $\theta = kx - \omega t$. Amplitude $2A \mid \cos(\delta / 2)\mid$.
Understand **Path difference**	In-phase sources, common wavelength: $\Delta r = m\lambda$ reinforces; $\Delta r = (m + \frac{1}{2})\lambda$ opposes. Equal arriving amplitudes needed for zero. Integer m.

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Recall **Standing wave**	$A\sin(kx-\omega t)+A\sin(kx+\omega t)=2A\sin(kx)\cos(\omega t)$. Equal opposite components; local amplitude $2A \mid \sin(kx)\mid$.
Recall **Nodes / antinodes**	For that sine form: nodes $x=n\lambda/2$; antinodes $x=(2n+1)\lambda/4$, integer n. A node is zero at all times.
Recall **Fixed-end modes**	$\lambda_n=2L/n$; $f_n=nv_{\mathrm{w}}/(2L)$, $n=1,2,\ldots$. Uniform string, common speed, fixed ends; n lobes, $n-1$ interior nodes.
Understand **Boundary response**	Transverse string displacement: fixed end inverts; ideal free end does not. Stationary linear boundary, steady source: frequency unchanged; $\lambda_2=v_2/f$.
Understand **Diffraction scale**	Opening width a: compare a/λ. Large ratio: narrower main spread; comparable/smaller ratio: broader spread. Spread and transmitted power differ.
Understand **Validation**	Check a signed value, crest or node, a nonzero antinode and endpoints. Refine sampling. Model agreement tests implementation; observations test physical suitability.

Consult · Python

Metres, seconds, radians. Tolerance below applies to these synthetic values.

```
import numpy as np
x = np.linspace(0, 2, 401)       # metres
A, lam, f, t = 0.003, 2.0, 1.0, 0.0
k, omega = 2*np.pi/lam, 2*np.pi*f
phase = k*x - omega*t
y1 = A*np.cos(phase)
y2 = A*np.cos(phase + np.pi)
y = y1 + y2
assert np.allclose(y, 0, rtol=0, atol=1e-12)
# Standing product: equal sine components
standing = 2*A*np.sin(k*x)*np.cos(omega*t)
```

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